

L8: δ -MAPS: Algorithm Overview

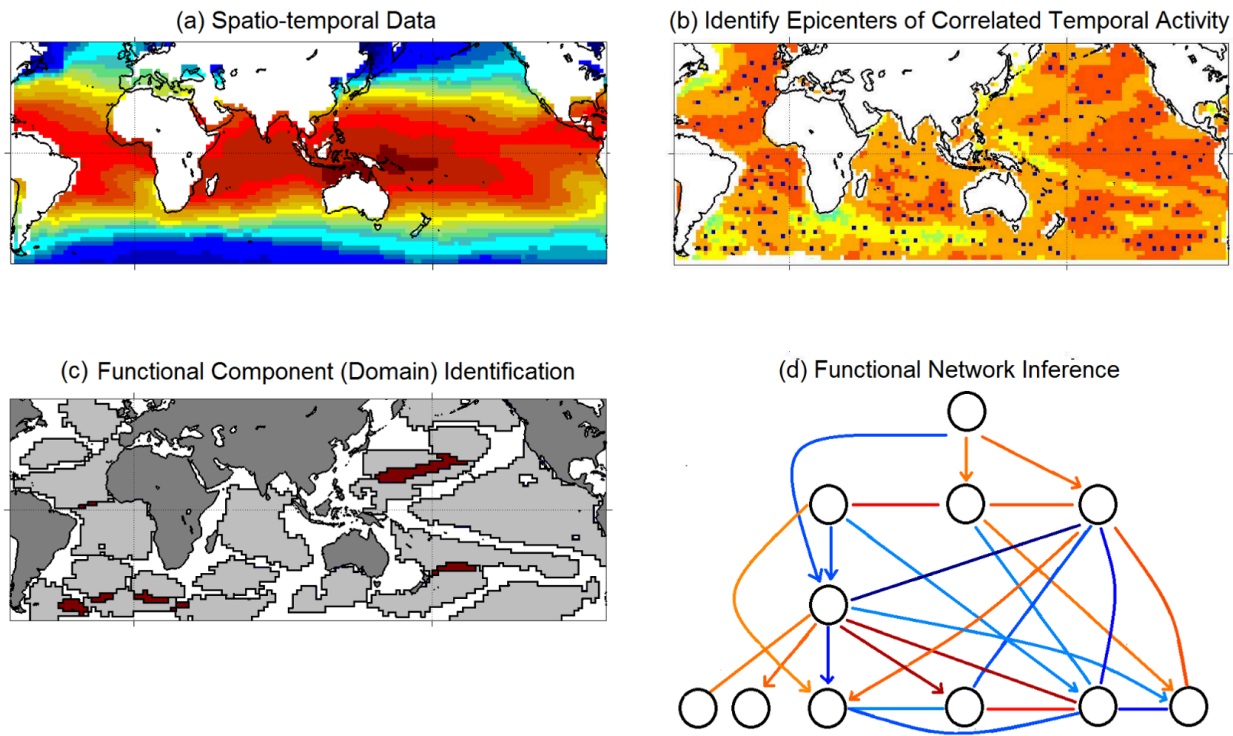


Figure from "[delta-MAPS: from spatio-temporal data to a weighted and lagged network between functional domains](https://link.springer.com/article/10.1007/s41109-018-0078-z)"
(<https://link.springer.com/article/10.1007/s41109-018-0078-z>) by Fountalis, I., Dovrolis, C., Bracco, A. et al., 2018

Here is a short overview of the Delta-MAPS algorithm:

The algorithm starts with a time series $x_i(t)$ for each point in space (*part-a shows the average of that time series for each grid point*).

Then, Delta-MAPS selects potential domain seeds. These points are local maxima of the spatial correlation field (*part-b, "epicenters of correlated temporal activity"*).

Then, the algorithm proceeds iteratively, identifying one domain at a time. The domain identification starts by selecting the next available seed of maximum spatial correlation. The algorithm then selects, in a greedy manner, as many points as possible around the chosen seed so that it forms a maximal spatially contiguous block of points that satisfy the homogeneity constraint δ (*part-c, "domain identification"*).

Finally, the algorithm identifies functional connections between domains, based on how strong their pairwise correlations are. These correlations can be positive (*red*) or negative (*blue*), and lagged (*part-d, "functional network inference"*).